Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. • **v2.7** • 16 August 2026 • supersedes the v2.6 PDF of the same date

> **What this revision does.** v2.4 put the critical path outside Track A for the first time: host it, then make it safe. Session 8 built the hosting half. **The schema now exists on a real database** (Supabase, eu-central-1, migrated and stamped), **tenant isolation is enforced by Postgres rather than by convention** (RLS forced on all 20 tables, behind a role that cannot bypass it), and **the API is a container image** at the path cd.yml has been waiting on since it was written — and **the API is deployed and serving** at watcher-api-lup7.onrender.com. What is left of Track B is a domain and the durable queue. **Remaining: ~34.25 engineering days.**

Where we stand today

Built and tested	Missing — nothing ships without these
417 passing tests, no DB or network needed 86 source files across 20 modules (125 with tests) 19 DB tables + 4 migrations, **applied to a live database** Every external system behind a swappable seam Eval runner + CI accuracy gate, 50 golden cases Receptionist vocabulary: 19 intents, data not code Prompt v3 — catalogue rendered from the vocabulary Channel-neutral envelope + chat/voice adapters Task state machine and the autonomy gate Typed config over every .env variable (A1) Anthropic + OpenAI on the LLM seam, caching on (A3) **NEW** • Engine + session scope, pooler-safe by default (A2) **NEW** • Composition root: the process starts and files a message end to end (A4) **NEW** • Conversation continuity: one task carried across turns, classifications recorded (A5) Outbound sender — the reply reaches the guest (A6) **NEW** • Supabase project, migrated, with the channel_configs row (B1) **NEW** • RLS forced on every table, per-transaction tenant GUC, cross-tenant test (B2) **NEW** • Container image; the project installs from pyproject (B3) **NEW** • **Deployed and serving** on Render, Frankfurt (B3)	**No DB connection — done (A2)** **No entrypoint — done (A4)** **No memory across turns — done (A5)** **No outbound — done (A6)** **No emergency detection** — emergency=False is still hardcoded (G3) **No slot extraction** — the model emits no slots, so a task fills only by handing off (2.x) **No database — done (B1)** **No RLS — done (B2)** **No Dockerfile — done (B3)** **No outbound credentials** — the deployed process warns at startup: it composes and records replies it cannot deliver **The database path is unproven** — SQLAlchemy connects lazily, so the pooler host is tested by the first message, not by startup No custom domain, no webhook subscription — nothing routes a guest to it yet (B4) No control page — one CSS file, no package.json No control-page API — ~25 endpoints, none written Knowledge — zero tables, zero rows Live availability — no read path to any PMS A measured prompt — the gate replays v2 fixtures, so 88% is not v3's number

What changed in this revision (v2.4 → v2.5)

Delivered	Consequences
B1, B2 and B3 delivered — 2.75 engineering days. Total 37 → **34.25** **Track B: 4.25 → 1.5 days** Tests 406 → **417** (+11). Migration 004 is the first one nothing autogenerated **watcher_app** — the application connects as a role that cannot bypass RLS; Supabase's postgres role can, which would have made the policies decorative Every tenant-facing adapter now takes a TenantScope rather than a SessionScope	**Tenant isolation is a property of the database** now, not of remembering the WHERE clause — the claim AGENTS.md calls non-negotiable is enforced and tested **Building the image found two defects the pinned test environment hides**: the intent vocabulary was missing from the wheel, and Starlette 1.0 removed add_event_handler **G3 is now the only thing between here and a number a guest can message** — B4 is half a day of DNS once a service exists **The service is live** — four deploys to get there, three of which failed on the config gate refusing to start without its secrets, which is the gate working **Tracks D, G, E and 3 are itemised** for the first time since v2.1, at unchanged track totals — 27 items that were four summary lines Unchanged: 2.4 is still the last item before a demo; 3.1 still blocked on P1; the scope guard still holds — no PDF handbook ingestion

Totals: ~34.25 engineering days remaining. At the observed 2–3 engineering days per working session that is **11–17 sessions**; at the six-day week these estimates assume, ~5.7 weeks of pure build — so plan **~7.5 weeks to sellable** and **~4 days to a real number that answers safely**. The critical path is now **G3 to make it safe, B4 to expose it, and 2.4** — G3 first, because the system already answers.

Track A — Make it run. [COMPLETE — ALL SIX DONE]

Configuration, a database, a model, an entrypoint, continuity and a sender. The track that turned a library into a running receptionist is finished; what follows is hosting it safely.

#	Work item	Urgency	Ease	Days	Status
A1	**Configuration layer**	DONE	Easy	0.5	**DONE** — core/config.py; every variable in .env.example typed, required per subsystem, placeholders read as absence, secrets as SecretStr
A2	**DB engine and session**	DONE	Easy	0.5	**DONE** — db/engine.py; engine, sessionmaker, session scope. Transaction-mode pooling assumed by default (NullPool + prepare_threshold=None), DATABASE_POOL_MODE opts out; a bare postgresql:// URI is rewritten onto psycopg 3; alembic connects the same way
A3	**Concrete LLM providers**	DONE	Moderate	1.5	**DONE** — Anthropic + OpenAI on the existing seam; system block marked cacheable; per-call usage reported, so cost is measurable
A4	**Composition root**	DONE	Moderate	1.0	**DONE** — apps/api/main.py, run as `uvicorn apps.api.main:create_application --factory`. Also the five database port implementations create_app needed and a process-level worker pool, so the webhook still answers before classification runs
A5	**Wire continuity + excise v1 filer**	DONE	Moderate	2.25	**DONE** — a ConversationStore in the message path: the inbound turn is recorded, the active task is loaded and saved back, the reply joins the transcript. process() is async; classifications is populated with measured latency and the prompt version; the rules/destinations threading is gone (D24), tables retained
A6	**Outbound sender**	DONE	Moderate	1.0	**DONE** — WhatsAppSender on the ChannelSender seam, quick replies rendered within the channel's limits, bounded retries. The credential fields moved behind channels/config.py and the last KNOWN_LEAKS entry is gone with them

> **What Track A leaves behind.** Items 2.1 and 2.2 were genuinely written — the tables, the repository, the task state machine, the reply path — and the line joining them never was. A5 wrote that line and A6 wrote the wire. The milestone “holds a conversation” is now true, and the next thing between this and a real guest is not another feature: it is **B1–B4 to host it and G3 to make it safe**.

Track B — Host it. [Supabase + Render]

Stack locked 15 August 2026. B2 must land before a second client exists. B1 grew one precondition this session — see its note.

#	Work item	Urgency	Ease	Days	Status
B1	**Supabase project and migrations**	DONE	Easy	0	**DONE** — watcher-prod in eu-central-1, migrations applied and stamped, one enabled channel_configs row. Its external_id is a placeholder until the Meta phone-number id replaces it — one UPDATE, in the deploy runbook
B2	**Row-Level Security**	DONE	Moderate	0	**DONE** — migration 004: enabled and forced on all 20 tables, policies on app.current_tenant, and a watcher_app role that cannot bypass them (Supabase's postgres role can, which is the trap). Verified cross-tenant on the live database
B3	**Dockerfile and Render service**	DONE	Easy	0	**DONE** — apps/api/Dockerfile, which activates the cd.yml image job, and the service is live in Frankfurt on the free plan. Sending is still degraded until WHATSAPP_ACCESS_TOKEN and WHATSAPP_PHONE_NUMBER_ID are set
B4	**Domain, TLS, webhook subscription**	HIGH	Easy	0.5	**NOT STARTED** — the platform needs a public HTTPS URL before anything arrives
B5	**Durable queue (Redis + arq)**	MED	Moderate	1.0	**NOT BUILT** — the in-process worker pool A4 ships loses in-flight classifications on every deploy. Persist-before-enqueue means a message is never lost, only its classification. Same seam, same consumer

> **Runs in parallel — start today.** P1 pick the first client (NOW, blocks 3.1) • P2 PMS sandbox keys (HIGH, 0.5d) • P3 file the platform business verification (HIGH, 1 hr) • P4 Graphify as a build aid (LOW, optional). Founder time, excluded from the total — and P2 and P3 are the two that make Track B's week land on time.

Track 2 — Make it a receptionist. [2.4, 2.7, 2.8 REMAIN]

What remains is knowledge, a measured prompt, and more than one property.

#	Work item	Urgency	Ease	Days	Status
2.1	**Conversations, tasks and slot filling**	HIGH	Hard	spent	**WIRED (A5)** — 2.0d spent here, the wiring priced in A5. Slot *extraction* is the one part still missing: the model emits no slots, so a task fills only by the clarifying-turn budget expiring. That is a prompt change and a golden-set change
2.2	**The reply path**	HIGH	Moderate	spent	**WIRED (A6)** — 1.5d spent here, the sender priced in A6
2.3	**Autonomy gate**	DONE	Easy	1.0	**DONE** — RECEPTIONIST_REPLY as the fourth RoutingAction
2.4	**Knowledge base**	HIGH	Moderate	2.0	**NEXT after Track A** — facts table with sensitivity flags, a real “I don't know” that fetches a human, verification codes
2.5	**Prompt v2 and rewrite the golden set**	DONE	Moderate	1.0	**DONE** — golden set 8 → 50 across all 19 intents
2.6	**Intent taxonomy unification**	DONE	Easy	0.5	**DONE** — 19 vocabulary-aligned intents
2.7	**Re-record the eval under prompt v3**	MED	Easy	0.5	**UNBLOCKED** — A3 landed; needs only a key. Add Franco-Arabic cases, and settle the two open numbers on the next page
2.8	**Many properties per client**	HIGH	Moderate	1.0	properties table, per-property fact scoping, message → property resolution

> **The eval is still not measuring the prompt.** The gate runs and passes, but it replays fixtures keyed by message text and recorded under prompt v2, so it reports 88% whatever the prompt says. Until 2.7 runs, treat 0.88 as v2's number. baseline.json says so in the file, and correctly records claude-haiku-4-5-20251001 as the model that produced those fixtures.
>
> **Tracks D, G, E and 3 are itemised below for the first time since v2.1.** Earlier revisions summarised them in four lines and pointed at an older PDF, which made half the remaining plan unreadable without a second document. The items are re-derived here from DESIGN-SPEC §8, the build-spec addendum and the current code — **track totals are unchanged** (D 13.75, G 5.5, E 4.5, 3 5.5), so the arithmetic on the previous page still reconciles. Where the numbering differs from v2.1, this supersedes it.

Track G — Receptionist guardrails 5.5d

Not new features. Every one of these enforces a rule that intents.yaml already states and schema.py already validates, and that nothing in apps/ honours — the vocabulary is obeyed when composing a reply and ignored when deciding whether to send one. G3 is first on the critical path.

#	Work item	Urgency	Ease	Days	Status
G3	**Emergency detection and the alert path**	NOW	Moderate	1.5	**worker.py hardcodes emergency=False**, at one named line with a comment. The vocabulary already declares the triggers and the alert; core/autonomy.py already takes an emergency flag and short-circuits everything on it. Missing: the detector, and who gets woken. **Answering raised the cost of this** — before A6 a gas leak was filed in silence; now it gets a confident note about maintenance
G1	**Sensitivity and disclosure gate**	HIGH	Moderate	1.5	A door code is not an ordinary fact. Facts carry sensitivity flags (2.4) and nothing yet refuses to say one to an unverified guest. Also the identity-verified flag on conversations, written and never read
G2	**Autonomy ceilings in the reply path**	HIGH	Easy	1.0	Money and owner matters must reach a human *before* confidence is consulted. The gate exists (2.3) and the vocabulary declares per-intent ceilings; what is missing is the enforcement that a ceiling cannot be raised by a confident model
G4	**Anti-loop and abuse controls**	MED	Moderate	1.5	The clarifying-turn budget bounds one task. It does not bound a guest who sends forty messages, a channel that redelivers, or two tasks that hand off to each other. Cheap to add now, expensive to retrofit after a bill arrives

Track D — The control page 13.75d

The largest remaining track, and the one whose shape is most often misread. **D2 is three backend days inside an item everyone reads as frontend work** — the views cannot be built against endpoints that do not exist. Views follow DESIGN-SPEC §8, re-ordered for the receptionist: what the product does now is hold conversations, not file messages.

#	Work item	Urgency	Ease	Days	Status
D0	**Frontend scaffold and design tokens**	MED	Easy	1.0	One CSS file and no package.json today. Tokens, type scale and the confidence chip come straight from DESIGN-SPEC §2–§6; the chip is the signature component and the bands are semantic, never reused
D1	**Auth and tenant binding**	MED	Moderate	1.25	Clerk (D2-a), one auth org to one tenant. This is where the RLS session GUC (B2) gets its value from an authenticated principal rather than from the message
D2	**The REST API behind the page**	HIGH	Hard	3.0	**~25 endpoints, none written.** Inbox list and detail, confirm/correct/route, conversations and turns, knowledge CRUD, sources, destinations, rules, eval. The hidden half of this track and the item most likely to be cut by accident
D3	**Inbox view**	HIGH	Hard	2.0	The triage queue and the product's centre of gravity. Two-pane desktop, single-column mobile, optimistic confirm. Auto-handled items still appear, marked, for audit
D4	**Conversations view**	HIGH	Moderate	1.5	New since v2.1 and it did not exist as a view when the track was first costed: the thread, its turns, the task in flight and its slots. Where a human sees *why* the receptionist asked what it asked
D5	**Knowledge view**	MED	Moderate	1.0	The editor for 2.4's facts, including the sensitivity flags G1 enforces. Without it, changing a check-in time is a database write
D6	**Sources view**	LOW	Easy	0.75	Opt-out model (addendum §4): a settings page listing threads with a mute toggle, plus the first-run exclusion pass. Deliberately not a workflow
D7	**Destinations and recipes**	MED	Moderate	1.0	Sheets/webhook config and field mapping. The tables and the engine survived D24 for exactly this: routing a message deliberately, by a human
D8	**Rules builder**	MED	Moderate	1.0	The condition/action builder over rules. SqlAlchemyRulesProvider and the evaluator are already written and already tested — this is the surface for them
D9	**Admin / eval dashboard**	LOW	Easy	0.75	Founder-only, role-gated, visually distinct: accuracy, per-language breakdown, calibration, confusion matrix, and per-tenant model spend
D10	**Arabic, RTL and the accessibility pass**	MED	Moderate	0.5	Not cosmetic for this market. Bidi text, mirrored layout, and the §10 baseline (focus order, contrast, target sizes) applied once across the views rather than per view

Track E — What makes it sellable 4.5d

None of this is visible to a guest and none of it can be added after the first paying tenant without a migration. Sequenced after P1 because onboarding shape depends on who the first client is.

#	Work item	Urgency	Ease	Days	Status
E1	**Tenant onboarding and per-tenant secrets**	MED	Moderate	1.25	Creating a tenant is currently an INSERT by hand. Includes envelope encryption for WABA tokens and destination credentials (addendum §3: never plaintext) — the reason channel_configs.config is the one table B2 grants a pre-tenant read on
E2	**Usage metering**	MED	Easy	0.75	usage_events exists, is indexed, and is written to by nothing. Messages, model tokens and ASR minutes per tenant per period — the input to both billing and the soft cap
E3	**Billing and plan limits**	MED	Moderate	1.0	Plans, the soft cap and what happens at it. The number itself is still a founder decision (DECISIONS.md, still open) — the mechanism is not
E4	**Observability**	HIGH	Easy	1.0	Structured logs with tenant and message ids, error tracking, and per-tenant model spend. Today a failed classification is a line in a Render log and nothing else. This is what makes a pilot debuggable at a distance
E5	**Retention, residency and subject deletion**	MED	Moderate	0.5	Addendum §14: a configurable retention default, hard deletion of raw bodies and media after it, and a “delete everything for this contact” operation. GCC regulated buyers ask for this in the first meeting

Track 3 — Integration and measurement 5.5d

Base360.ai is ruled out as a partner: their product is substantially ours and their client base is closed to us. Hostaway, Guesty and Cloudbeds all publish APIs, so the capability is available without the strategic cost.

#	Work item	Urgency	Ease	Days	Status
3.1	**PropertySystemPort and the first adapter**	MED	Moderate	2.5	**Build the port, not a Hostaway integration** — the vendor-neutral routes and schemas already landed (PR #14). Blocked on P1: which PMS the first client runs decides which adapter gets written. Cache facts; never cache availability
3.2	**End to end on a real number, then measure**	HIGH	Moderate	1.0	The point at which the eval number stops being a replay of recorded fixtures and starts being this product's accuracy. Needs B4, G3 and the send credentials first
3.3	**Live availability and write-back**	MED	Hard	2.0	Reading a calendar is half of it; holding a booking is the half that makes the receptionist worth paying for, and the half that must never act on a stale cache

New this revision: what A5 and A6 settled, and what they left open

Continuity	Every classified message now runs against a **ConversationStore**: the inbound turn is recorded, the job already in flight is loaded, and the updated job and the reply are written back. A receptionist constructed without a store is **refused** — one that forgets the previous turn looks like it works, which is the exact failure this item existed to remove. ConversationRepository, written in item 2.1, is finally called by something other than its own tests
The clarifying-turn budget	Continuity introduces a failure mode that no-continuity does not: a task that cannot make progress no longer fails, it **loops**. The vocabulary has declared max_clarifying_turns = 3 and on_max_turns = handoff_to_human since item 0.3 and nothing read them. They are read now. The guard applies to asking, never to acting: a task with everything it needs still completes on the last turn of the budget
The v1 filer	The orchestrator used to answer an inbound message twice — auto-route it by rule or band, *and* reply to it. Routing to a spreadsheet was v1's answer; the receptionist is v2's. **Rules and destinations are gone from the message path (D24) and both tables are retained**, because deliberate routing by a human is a control-page feature and this is the evaluator it will use. AUTO_ROUTE went with them: there is nothing to route to
classifications, at last	The table sat empty for four sessions because two of its columns — latency_ms and prompt_version — had no honest source. The classifier reports them now rather than the writer inventing them: **latency spans the whole tiered policy**, retries and escalation included, because that is what the guest waited for. model_used, which used to be handed to the inbox writer and dropped, lives on that row, and inbox_items points at it
The wire	WhatsAppSender posts to the versioned Graph endpoint for the number we send as. The three-button cap and the 20-character title limit are applied there and nowhere else, and neither raises. **A failed send is logged and reported, never raised** — by then the message is classified, the reply recorded and the decision filed, and a transient 502 must not undo all of it. The reply is recorded *before* it is sent, deliberately
Item 1.1, closed	The channel credential fields moved from core/config.py to **channels/config.py**, which Settings extends: one object still reads one environment, but the knowledge of what a send needs sits with the channel. **KNOWN_LEAKS is now empty** — no core file names a channel — and the machinery stays, because an empty allowlist is the strongest form of that test and a phone line is next
Still not written	**Slot extraction.** The receptionist receives an empty slot dict because ClassificationResult has no slots field; adding one is a prompt change and a golden-set change, which is item 2.x. The consequence is bounded and honest: a task fills only by the budget expiring and then fetches a person. **Emergency detection is still hardcoded to False** — that is G3, and it is the last safety gap

Two numbers nobody has measured, both cheap to settle during 2.7

1. The prompt's real token count	~5k is a characters÷4 estimate. Haiku 4.5 will not cache a prefix below **4,096 tokens**, and the cheap tier is where caching pays for itself. The estimate clears the floor by ~30%, and the true count is probably higher (Arabic and Franco-Arabic tokenize worse than the approximation), so caching almost certainly activates — but if the vocabulary shrinks it stops silently: no error, just a larger bill. Sonnet 5's floor is 1,024, so escalation is not exposed either way
2. Cost per message	Still unmeasured, but **measurable**: TokenUsage reports cached versus fresh input tokens per call, normalised across both providers. Measure before quoting a price

How the total reconciles

A 0 + B 1.5 + D 13.75 + E 4.5 + G 5.5 + (2.4, 2.7, 2.8) 3.5 + (3.1, 3.2, 3.3) 5.5 = **34.25**

Rows 2.1 and 2.2 show “spent” rather than a number because their days are already delivered and their wiring was priced in A5 and A6, now built — adding them would double-count 3.5 days. P1–P4 is excluded: founder time, not engineering. Counting scope loosely is what produced the 5.75 figure that v2.1 replaced, so the arithmetic is stated rather than implied.

The dates

Milestone	v2.5 said	Now	Status
M1 — answers a real message, safely	~4.25 days	**~4 days**	**NEXT** — G3 + B4 + 2.4. It is deployed and it answers; the emergency path is what “safely” still depends on
M2 — you can watch and correct it	+13.75 days	+13.75 days	PENDING — Track D, six receptionist views
M3 — sellable	+16.5 days	+16.5 days	PENDING — Tracks E, F and the rest of G. Blocked on P1
Phone answering	after M3	after M3	PENDING — the speech-to-text seam already exists

What would actually move these dates

The risks v2.3 named, re-scored against what session 7 learned.

Supabase connection pooling	**Downgraded from imminent to unmeasured.** A2 assumes the transaction pooler and is safe there by default. What is not known is the cost of that safety — NullPool trades a handshake per checkout for correctness. Measure it during B1/B3 and flip DATABASE_POOL_MODE if session mode wins; do not settle it in production at 2am
A missing channel_configs row	**NEW.** The first deployed webhook will 500 on every message until B1 inserts one row per endpoint. It is a one-line fix and a genuinely confusing hour if nobody wrote it down, which is why it is in B1's status, the README and the spec
RLS retrofitted late	Unchanged. Adding row-level security after application queries exist surfaces as silent empty result sets rather than errors. B2 sits before the second tenant for that reason
D2 is invisible in its own track title	Unchanged, and still the most likely accidental cut. Three of Track D's days are backend
Cost per message	Unmeasured, not unknown. A3 made it measurable and turned prompt caching on. Measure before quoting a price
PMS API access	Unchanged. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 today
Three roadmap artifacts disagree	**CLOSED.** docs/make_roadmap.py generates this document into docs/Watcher_v2_Roadmap.pdf. Edit the generator and re-run it; do not hand-write another version
A receptionist that answers but cannot recognise an emergency	**NEW, and the most serious item on this page.** Until A6 a gas leak was filed in silence; now it gets a polite, confident reply about maintenance. Answering raises the cost of emergency=False rather than lowering it, which is why **G3 moves ahead of 2.4** for any deployment a real guest can reach
Two messages arriving at once	**NEW, bounded.** Two turns in one thread classified concurrently can race and open two conversations. Per-message ordering is a property of the queue, and the queue is an in-process pool until **B5**, where ordered per-conversation delivery belongs. Recorded rather than papered over with a lock that would not survive a second process

> **If you do only one thing next session: B1, then G3.** Not 2.4, and not D. The receptionist now answers — on a laptop, to a number nobody can message. B1–B4 is 3.25 days and ends with a public HTTPS URL the platform can deliver to (and B1 must insert the channel_configs row, or every message 500s). **G3 is 1.5 days and it is what makes answering safe**: emergency=False is hardcoded, so a gas leak now gets a confident reply about maintenance rather than a person. Answering made that worse, not better.

Session log

Session	Delivered	Tests	Days
1	PR #13 — items 1.1, 1.3, 2.1, 2.2, 2.3	228 → 248	6.5
2	PR #14 and #15 — items 2.5, 2.6	248 → 259	1.5
3	PR #16 — prompt v3, unplanned	259 → 277	~0.5
4	Code audit plus a design review. No application code changed. Five unpriced tracks added; 2.1/2.2 downgraded to part-built; emergency detection found unwired; the control page rebuilt around six receptionist views. Decisions D14–D26	277	—
5	PR #17 — items A1, A3. Config layer over all ~20 env vars; Anthropic + OpenAI providers with prompt caching and per-call usage; D8-a re-pinned to the Claude 5 family	277 → 325	2.0
6	PR #18 — items A2, A4. Engine and session scope, pooler-safe by default; the composition root and the five database port implementations it needed; a process-level worker pool so the webhook still answers before classification; alembic aligned to the same connection path	325 → 375	1.5
7	Items A5, A6 — Track A complete. Conversation and task continuity in the message path; the clarifying-turn budget honoured; classifications populated with measured telemetry; process() async; the v1 rules/destinations filer excised (D24); the outbound sender, and the channel credentials moved behind channels/, which emptied KNOWN_LEAKS and closed item 1.1	375 → 406	3.25
8	**Items B1, B2, B3.** Supabase project provisioned, migrated and stamped, with the channel_configs row tenant resolution raises without; migration 004 — RLS enabled and forced on all 20 tables behind a watcher_app role that cannot bypass it, a per-transaction tenant GUC carried by every adapter, and one narrow SELECT-only exception for the endpoint lookup that runs before a tenant is known; the container image, whose first build exposed a missing package data file and a removed Starlette API; deployed to Render, Frankfurt, and serving	**406 → 417**	**2.75**

Cumulative: 18 engineering days delivered. ~34.25 remaining.